



Sheryians
Coding School

POWER BI

Learn How Data Analysts Work And How Power BI Turns Raw
Data Into Insights.

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Power BI

DISCOVER DATA ANALYSIS

Roles In Data (Who Does What?)

In The Data World, Different Roles Exist:

- **Data Analyst**
Analyzes Data, Creates Reports & Dashboards, Finds Insights.
- **Data Engineer**
Builds Pipelines, Prepares Large Data Systems.
- **Data Scientist**
Uses ML & Statistics For Predictions.

Power BI Is Mainly Used By Data Analysts.



Tasks Of A Data Analyst

A Data Analyst's Work Follows A Clear Cycle:

Data Analyst Workflow

1. Prepare – Clean & Shape Data
2. Model – Create Relationships & Calculations
3. Visualize – Create Charts & Dashboards
4. Analyze – Find Trends, Patterns, Insights
5. Manage – Share, Refresh & Maintain Reports

This Cycle Repeats Again & Again In Real Jobs.



Power BI Ecosystem Overview

Power BI Has Three Main Parts:

Power BI Desktop

- Used To Build Reports
- Data Cleaning + Modeling + Visuals
- Free Tool

Power BI Service

- Cloud Platform
- Share Reports & Dashboards
- Schedule Data Refresh
 - Collaboration



Power BI Mobile

- View Dashboards On Phone
- Read-Only Access
- Used By Managers

Data Analyst Workflow & Business Use Cases

Power BI Has Three Power BI Is Used In:

- Sales Analysis
- Finance Reports
- HR Dashboards
- Marketing Performance
- Operations Tracking Main Parts:



Example:

Monthly Sales Report For Management

Employee Attrition Dashboard For HR

GET STARTED WITH POWER BI

Introduction To Power BI (Desktop, Service, Mobile)

Power BI Helps You:

- Connect Data From Many Sources
- Clean Messy Data
- Build Interactive Dashboards
- Share Insights With Others



Installation, Setup & Interface Overview

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Installation

- Download Power BI Desktop From Microsoft
- Works On Windows



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Interface Overview (Desktop)

Main Sections:

- Ribbon – Home, Transform, Modeling, View
- Report View – Create Visuals
- Data View – View Tables
- Model View – Relationships
- Fields Pane – Tables & Columns
- Visuals Pane – Charts & Graphs



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Data Analyst Workflow In Power BI

Step	Tool Used
Prepare	Power Query
Model	Power Pivot
Visualize	Power View
Analyze	DAX
Share	Power BI Service

This Is The Core Structure Of Power BI Learning.



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Connecting Data Sources

Power BI Supports Many Data Sources:

- Excel
- CSV / Text Files
- Web URLs
- SQL Server
- SharePoint
- JSON
- PDF

Real Companies Use Multiple Sources Together.



Basic Transformations In Power Query

Power Query Is Used To Clean Data.

Common Actions:

- Remove Unwanted Rows
- Remove Blank Rows
- Split Columns (Name, Date, Etc.)
- Fill Down Values
- Replace Values

These Steps Are Recorded Automatically.



Data Types & Column Management

Data Types

Examples:

- Text
- Number
- Decimal
- Date
- Time

Wrong Data Type = Wrong Result



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Rename Columns

- Keep Names Simple & Meaningful
- Avoid Spaces & Special Characters

Detect Data Type Errors

- Errors Appear When Data Doesn't Match Type
- Fix In Power Query



Semantic Model Vs Visual Layer

Semantic Model

- Tables
- Relationships
- Measures
- Business Logic

Brain Of Power BI

Visual Layer

- Charts
- Tables
- Cards
- Slicers

Face Of Power BI

Both Must Work Together.



Power BI Building Blocks

Dataset

- Cleaned & Modeled Data

Report

- Collection Of Visuals
- Multiple Pages

Dashboard

- Single-Page View (Service Only)
- Pin Visuals From Reports

Workspace

- Collaboration Space
- Used By Teams



Power BI In Microsoft Fabric Ecosystem

Power BI Is Part Of Microsoft Fabric, Which Includes:

- Data Engineering
- Data Warehousing
- Data Science
- Real-Time Analytics

Power BI Is The Visual & Reporting Layer.



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CASE STUDY

Cleaning Messy Excel Data Into Report-Ready Format

Problem:

- Blank Rows
- Wrong Data Types
- Mixed Columns
- Duplicate Records

Solution Using Power Query:

1. Import Excel File
2. Remove Blank Rows
3. Split Merged Columns
4. Fix Data Types
5. Rename Columns
6. Load Clean Data To Report

Result: Clean, Reliable, Report-Ready Data

